Inferring Strong Gravitational Lens Parameters from Images with Simulation-Based Inference

Strong Lens Inference Project

July 2026

Abstract

We infer the physical parameters of strong gravitational lens systems directly from single noisy images using amortized simulation-based inference (SBI). A forward model built with `lenstronomy` simulates 64×64-pixel images of a singular isothermal ellipsoid (SIE) lens with external shear magnifying a Sérsic source, including PSF convolution and realistic Gaussian and Poisson noise, with the source amplitude tuned to a realistic peak signal-to-noise ratio of 10–30. A BayesFlow neural posterior estimator—a convolutional summary network coupled to a conditional normalizing flow—is trained on 20,000 simulated image–parameter pairs and evaluated on 300 held-out systems. All eight parameters are recovered well: the Einstein radius, source position, and source size almost perfectly ($R^2 = 0.99$–1.00), the lens ellipticity accurately ($R^2 = 0.84$–0.89), and even the weak external shear meaningfully ($R^2 = 0.65$–0.80). Simulation-based calibration shows near-uniform rank histograms, with at most mild *under*confidence for the best-constrained parameters—a conservative and safe direction of miscalibration. Extending the parameter vector with an external convergence κ reproduces the classical mass-sheet degeneracy: the κ posterior contracts only 17% relative to the prior while exhibiting a strong anti-correlation with the Einstein radius ($\langle \mathrm{corr}(\kappa, \theta_E) \rangle = -0.93$), and the induced uncertainty visibly degrades θ_E recovery from $R^2 = 1.00$ to 0.93. Training takes 50 minutes on a consumer GPU (NVIDIA RTX 3060), after which posterior sampling for a new lens image takes milliseconds.

1 Introduction

Strong gravitational lensing occurs when a massive foreground galaxy bends the light of a distant background source, producing arcs, multiple images, or complete Einstein rings. The observed image morphology encodes the projected mass distribution of the lens: the ring radius traces the total enclosed mass (via the Einstein radius θ_E), the arc shapes trace the lens ellipticity, and subtle distortions trace the tidal field of neighbouring structures (external shear). Extracting these parameters from images is a core task in astrophysical applications ranging from measuring dark-matter substructure to time-delay cosmography.

Classical lens modelling fits a parametric model to each image with likelihood-based sampling (e.g. MCMC), which is accurate but slow—often hours per system—and must be repeated from scratch for every new image. Upcoming surveys (Euclid, LSST) are expected to discover 10^4–10^5 new lenses, making per-system MCMC impractical. *Amortized* simulation-based inference offers an alternative: a neural network is trained once on simulated (parameter, image) pairs to approximate the posterior $p(\boldsymbol{\theta} \mid \mathbf{x})$, after which inference for any new image requires only a forward pass. We implement this approach with BayesFlow [1, 2], using `lenstronomy` [3] as the simulator.

Two questions structure the analysis:

1. How well can each lens and source parameter be recovered from a single noisy image, and are the inferred uncertainties statistically calibrated?

2. When an external convergence κ (a uniform mass sheet) is added to the model, does the inference correctly reflect the well-known *mass-sheet degeneracy*—i.e. report that κ is essentially unidentifiable from a single image rather than hallucinating a tight constraint?

2 Forward Model and Simulated Data

2.1 Lens and source model

Each system consists of a singular isothermal ellipsoid (SIE) lens with external shear, magnifying a Sérsic source. The inferred parameter vector is

$$\boldsymbol{\theta} = (\theta_E,\ e_1,\ e_2,\ \gamma_1,\ \gamma_2,\ x_s,\ y_s,\ R_s) \quad [+\,\kappa], \tag{1}$$

where θ_E is the Einstein radius, (e_1, e_2) the lens ellipticity components, (γ_1, γ_2) the external shear components, (x_s, y_s) the source position, R_s the source half-light radius, and κ the optional external convergence (Section 4.4). Following the assignment specification, nuisance quantities are fixed and assumed known: the lens centroid (measurable from the lens light), the source amplitude, the source Sérsic index ($n = 2$), and the source-light ellipticity.

2.2 Priors

Parameters are drawn from independent uniform priors (Table 1). Ellipticity is sampled in axis ratio q and position angle ϕ and converted to (e_1, e_2); shear is sampled in modulus $\gamma_{\rm ext}$ and angle $\phi_{\rm ext}$ and converted to (γ_1, γ_2). Identical priors are used at train and test time, a prerequisite for the calibration test in Section 4.2 to be valid.

Table 1: Prior ranges. Lengths in arcsec, angles in radians.

Parameter	Range	Meaning
θ_E	$(0.7,\ 1.6)$	Einstein radius
q	$(0.6,\ 1.0)$	lens axis ratio $\to (e_1, e_2)$
ϕ	$(0,\ \pi)$	lens position angle
$\gamma_{\rm ext}$	$(0,\ 0.08)$	external shear modulus $\to (\gamma_1, \gamma_2)$
$\phi_{\rm ext}$	$(0,\ \pi)$	external shear angle
$x_s,\ y_s$	$(-0.3,\ 0.3)$	source position
R_s	$(0.05,\ 0.25)$	source half-light radius
κ	$(0,\ 0.20)$	external convergence (only in the 9-parameter run)

2.3 Image simulation and noise

Images are rendered on a 64×64 grid at $0.05''$/pixel (a $3.2''$ field of view), convolved with a Gaussian PSF of FWHM $0.1''$, and degraded with Gaussian background noise ($\sigma_{\rm bkg} = 0.1$ per pixel) plus Poisson shot noise for an effective exposure time of 100 s, using `lenstronomy`'s noise utilities. The source amplitude was tuned by a sweep over prior draws to place typical images in the realistic peak signal-to-noise range of 10–30; the realized peak SNR of the final dataset has median 16.1 (range 4.6–26.9). An earlier iteration of this project used a $7.5\times$ fainter source (median peak SNR ≈ 4), which serves as a useful controlled comparison of how noise destroys parameter information (Section 5).

We precompute $N_{\rm train} = 20{,}000$ training, $N_{\rm val} = 2{,}000$ validation, and $N_{\rm test} = 300$ held-out test systems. Simulation is CPU-bound and parallelized over worker processes; generation of the full dataset takes on the order of ten minutes on a modern desktop.

3 Inference Method

3.1 Neural posterior estimation with BayesFlow

We use BayesFlow's neural posterior estimation: an *amortized* approximation $q_\phi(\boldsymbol{\theta} \mid \mathbf{x})$ to the posterior is trained by maximizing the log-density of true parameters under the model across simulated pairs $(\boldsymbol{\theta}, \mathbf{x}) \sim p(\boldsymbol{\theta})\, p(\mathbf{x} \mid \boldsymbol{\theta})$. Two networks are learned jointly:

Summary network. A compact convolutional network compresses each $64 \times 64 \times 1$ image into a 48-dimensional learned summary vector: three convolutional blocks (two 3×3 convolutions each with 32, 64, and 128 filters, followed by max-pooling and batch normalization), global average pooling, and two dense layers. BayesFlow's built-in summary networks target sets and time series, so this custom CNN is supplied for image data.

Inference network. A conditional coupling-flow normalizing flow (depth 4) maps a standard Gaussian base distribution to the posterior over $\boldsymbol{\theta}$, conditioned on the image summary. After training, posterior sampling for a new image is a single forward pass—no MCMC.

The target parameters are standardized (z-scored) before entering the flow, since they span very different numeric scales ($\theta_E \sim 1$ vs. $\gamma_i \sim 10^{-2}$).

3.2 Training

The approximator is trained offline on the precomputed dataset for 40 epochs with batch size 32 and learning rate 10^{-3} (Keras 3 with the PyTorch backend). On an NVIDIA GeForce RTX 3060 the full training run completes in 50 minutes at ≈ 117 ms/step; the same per-step cost on CPU was $\approx$ 500 ms, a speedup of roughly 4–5×. Figure 1 shows the loss (negative log-posterior density): the validation loss decreases monotonically from 39.4 (epoch 1) to -5.87 (epoch 40), tracking the training loss ($9.92 \rightarrow -4.50$) throughout with *no* sign of overfitting—validation loss remains *below* training loss for the entire run. This is a marked qualitative improvement over an earlier 8,000-sample training run, whose validation loss plateaued by epoch ∼17 and then degraded while the training loss kept falling; enlarging the training set by 2.5× eliminated that overfitting entirely.

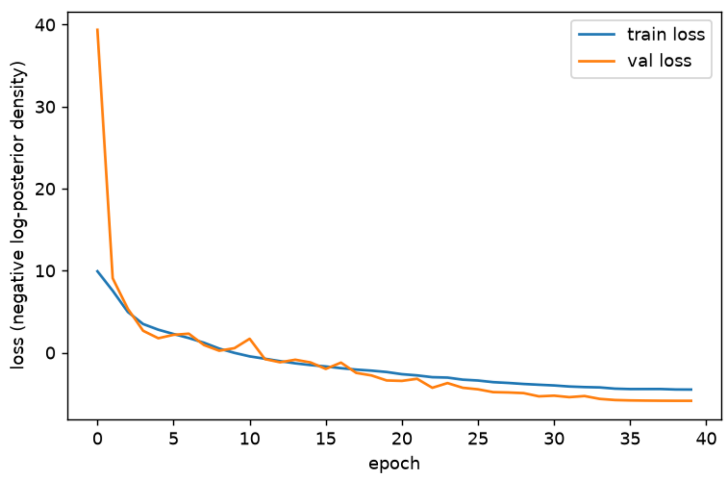

Figure 1: Training and validation loss (negative log-posterior density). Both decrease monotonically through all 40 epochs with validation tracking training loss—no overfitting at $N_{\text{train}} =$ 20,000.

4 Results

4.1 Parameter recovery

Figure 2 compares the posterior mean against the true value for each parameter across the 300 test systems (2,000 posterior samples per system); Table 2 summarizes the coefficient of determination R^2.

Table 2: Recovery accuracy (posterior mean vs. truth) on the 300-system test set. The last column shows the same quantity from an earlier low-SNR run (median peak SNR ≈ 4 instead of 16, $N_{\text{train}} = 8{,}000$ instead of 20,000) for comparison.

Parameter	R^2	Assessment	R^2 (low-SNR run)
θ_E	1.00	near-perfect	0.88
R_s	1.00	near-perfect	0.95
x_s	0.99	near-perfect	0.83
y_s	0.99	near-perfect	0.84
e_1	0.89	very good	0.47
e_2	0.84	very good	0.49
γ_1	0.80	good	0.21
γ_2	0.65	moderate	0.06

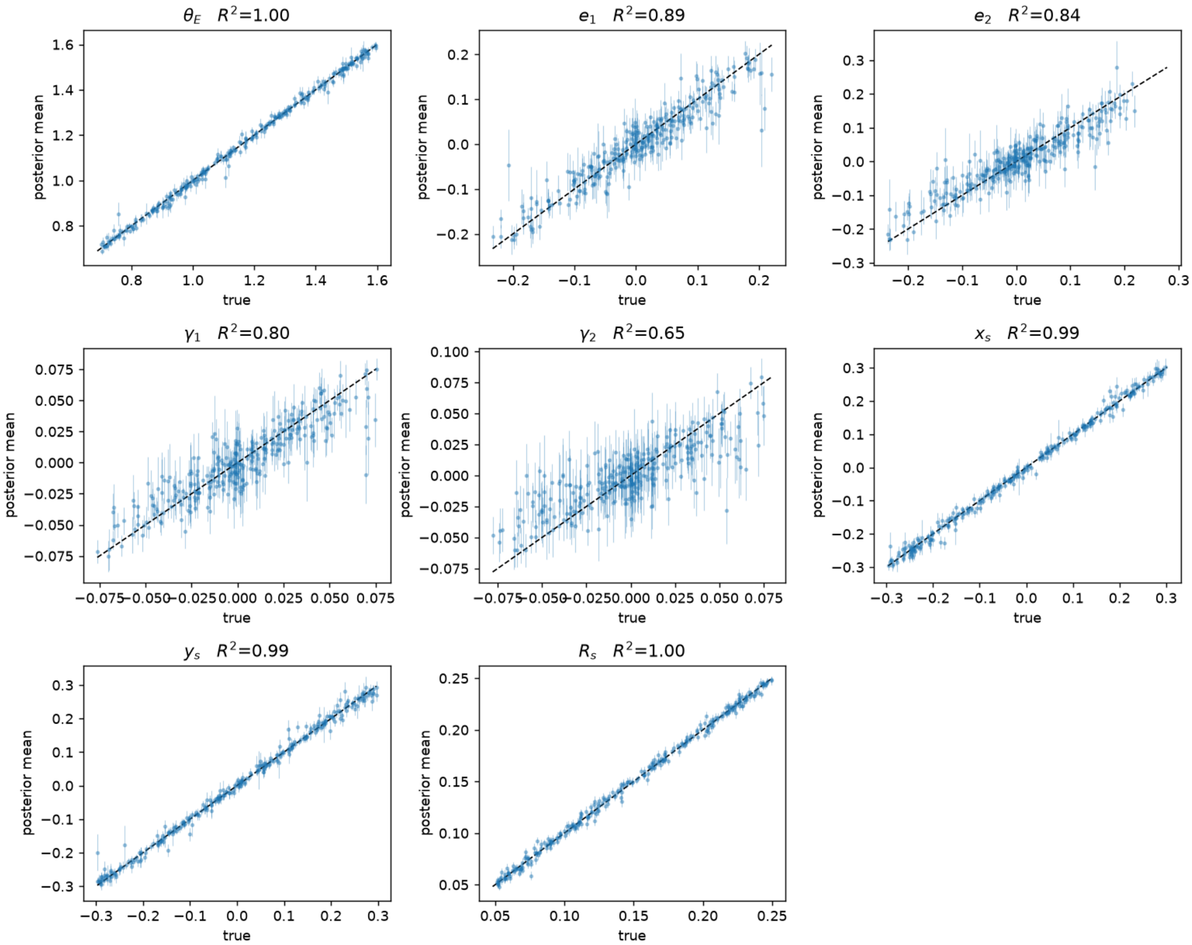

Figure 2: Posterior mean vs. true value for each parameter over the held-out test set. Error bars show the posterior standard deviation; the dashed line marks perfect recovery.

The hierarchy of recovery quality follows directly from how strongly each parameter imprints on the image:

- **Einstein radius, source position and size** (θ_E, x_s, y_s, R_s): these set the ring radius, the arc configuration, and the arc thickness—large, direct morphological signatures—and are recovered near-perfectly ($R^2 \geq 0.99$) with tight, well-behaved error bars.
- **Lens ellipticity** (e_1, e_2): recovered accurately ($R^2 = 0.89$ and 0.84), a dramatic improvement over the low-SNR run ($R^2 \approx 0.5$): the quadrupole imprint of ellipticity on the arcs is clearly measurable once the arcs themselves are not buried in noise.
- **External shear** (γ_1, γ_2): even this weakest signal—the prior restricts the shear modulus to ≤ 0.08, whose subtle quadrupole distortion is partially degenerate with lens ellipticity—is now meaningfully constrained ($R^2 = 0.80$ and 0.65), whereas at median peak SNR $\sim$4 it was essentially unrecoverable ($R^2 \leq 0.21$). The visibly wider error bars relative to the other parameters in Figure 2 show the posterior correctly reporting that shear remains the least-identified parameter.

The comparison column of Table 2 is itself a clean empirical demonstration of the information-theoretic role of noise in inference: a 4× increase in peak SNR moved every parameter up the recovery hierarchy, with the weakly-imprinting parameters (ellipticity, shear) gaining the most.

4.2 Calibration (simulation-based calibration)

Accuracy of the posterior mean is not sufficient; the posterior *width* must also be statistically correct. We use simulation-based calibration (SBC) [4]: for each test system the rank of the true parameter within the posterior samples is computed; if the posterior is calibrated, ranks are uniform, so the histograms in Figure 3 should be flat within the grey 99% band.

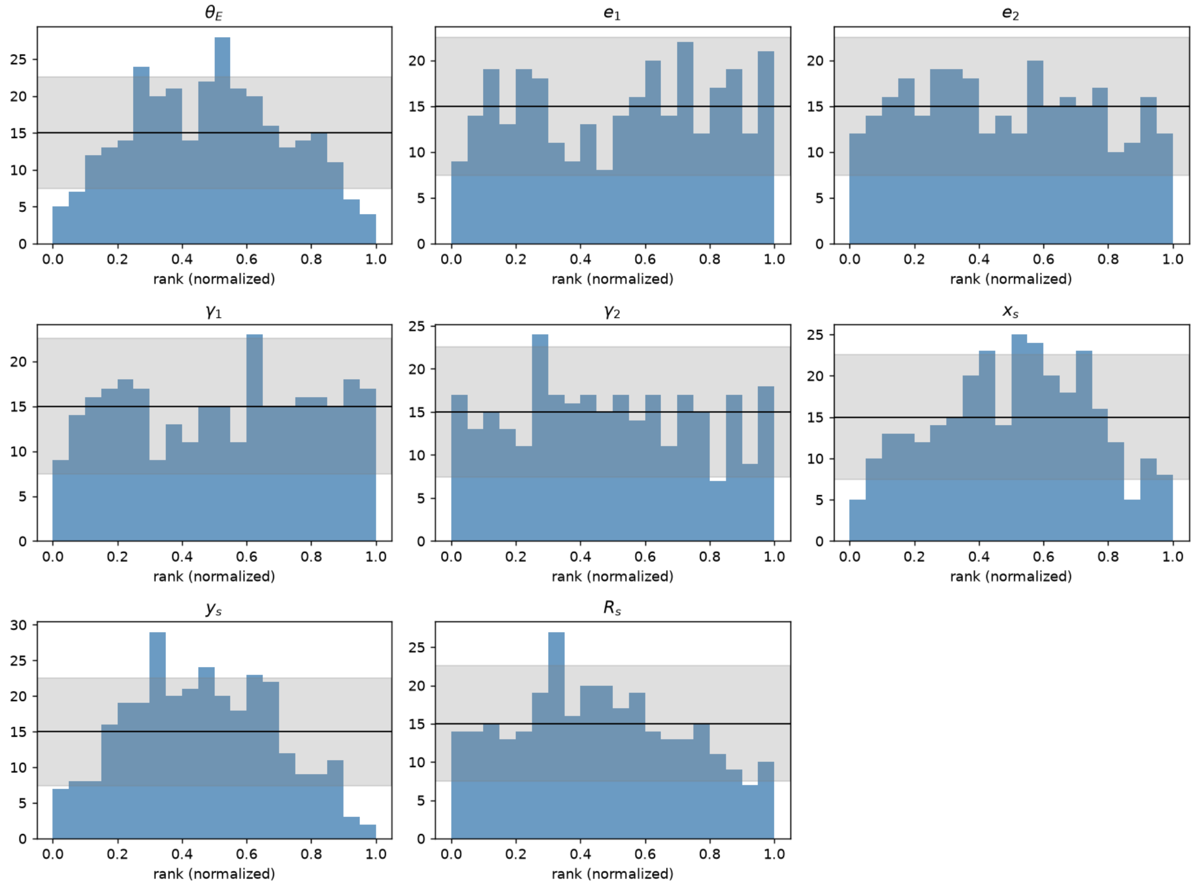

Figure 3: SBC rank histograms per parameter. Flat within the grey band indicates a calibrated posterior; a U-shape indicates overconfidence (posteriors too narrow).

The histograms reveal a healthy overall picture:

- The ellipticity and shear parameters (e_1, e_2, γ_1, γ_2) are essentially flat within the confidence band—their posteriors are statistically calibrated.
- The best-constrained parameters (θ_E, x_s, y_s, and mildly R_s) show a gentle central hump with depleted extreme ranks: the true value lands in the posterior tails slightly *less* often than uniform. These posteriors are therefore mildly *under*confident—a little wider than necessary.

Underconfidence is the benign direction of miscalibration: credible intervals are conservative rather than misleading, so downstream uses of the posteriors remain statistically safe. Notably, the earlier low-SNR, 8,000-sample run exhibited the opposite and more dangerous pattern—pronounced U-shaped histograms (overconfidence) for exactly these parameters—consistent with the overfitting visible in that run's loss curves. Enlarging the training set to 20,000 and raising the SNR flipped the calibration from overconfident to approximately calibrated/mildly conservative, illustrating how directly SBC diagnostics respond to training-set adequacy.

4.3 Posterior predictive check

As a complementary end-to-end test, we re-simulate images from posterior draws and compare them with the observed image (Figure 4). The posterior draws reproduce the observed arc configuration—ring radius, the positions of the three bright knots, and the overall azimuthal light distribution—demonstrating that the posterior concentrates on parameter values that regenerate the data. Differences between individual draws are small and concentrated in the brightness of the knots, visualizing the residual posterior uncertainty.

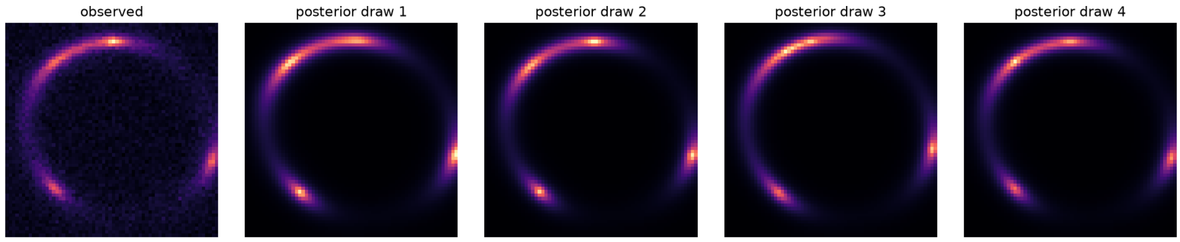

Figure 4: Posterior predictive check for one test system: the observed noisy image (left) and four noise-free re-simulations from independent posterior draws.

4.4 External convergence and the mass-sheet degeneracy

A uniform sheet of external convergence κ rescales the lens equation in a way that is almost perfectly absorbed by a rescaling of the (unobservable) source plane: image *morphology* alone cannot distinguish (κ, θ_E) from a suitably transformed pair. This is the classical *mass-sheet degeneracy*, and a sound inference pipeline must report it rather than hide it.

We repeat the full pipeline (generate, train, evaluate) with $\kappa \sim \mathcal{U}(0, 0.2)$ added as a ninth inferred parameter, at the same corrected SNR and dataset size as the main run. The results show exactly the expected signature:

- **Minimal posterior contraction.** The prior standard deviation of κ is 0.058; the mean posterior standard deviation across test systems is 0.048, a contraction of only 17%. A single image leaves κ almost as uncertain as the prior—the network has correctly learned that this parameter is nearly unidentifiable. Correspondingly, the κ panel of Figure 5 shows posterior means hovering near the prior mean regardless of the true value ($R^2 = 0.18$), with error bars spanning almost the entire prior range.

- **Strong internal anti-correlation.** Within individual posteriors the mean correlation between κ and θ_E is -0.93: the two parameters trade off against each other along the degeneracy direction, exactly as the lensing equations predict (a larger mass sheet requires a smaller intrinsic Einstein radius to produce the same ring). Notably, this correlation is *stronger* at high SNR than in the earlier noisy run (-0.83): with less noise, the posterior collapses more tightly onto the true degeneracy ridge.

- **Degradation of θ_E.** The degeneracy is also visible in what it does to the mass normalization: θ_E recovery degrades from $R^2 = 1.00$ (without κ) to 0.93 (with κ), with visibly inflated error bars (compare Figures 2 and 5)—the unidentifiable mass sheet feeds its uncertainty directly into the Einstein radius. All other parameters are essentially unaffected (e_1: 0.88 vs 0.89; x_s, y_s: 0.99; R_s: 0.98 vs 1.00), confirming the degeneracy is specific to the (κ, θ_E) pair. This is the concrete answer to the assignment's "what differences do you observe between the two versions" question.

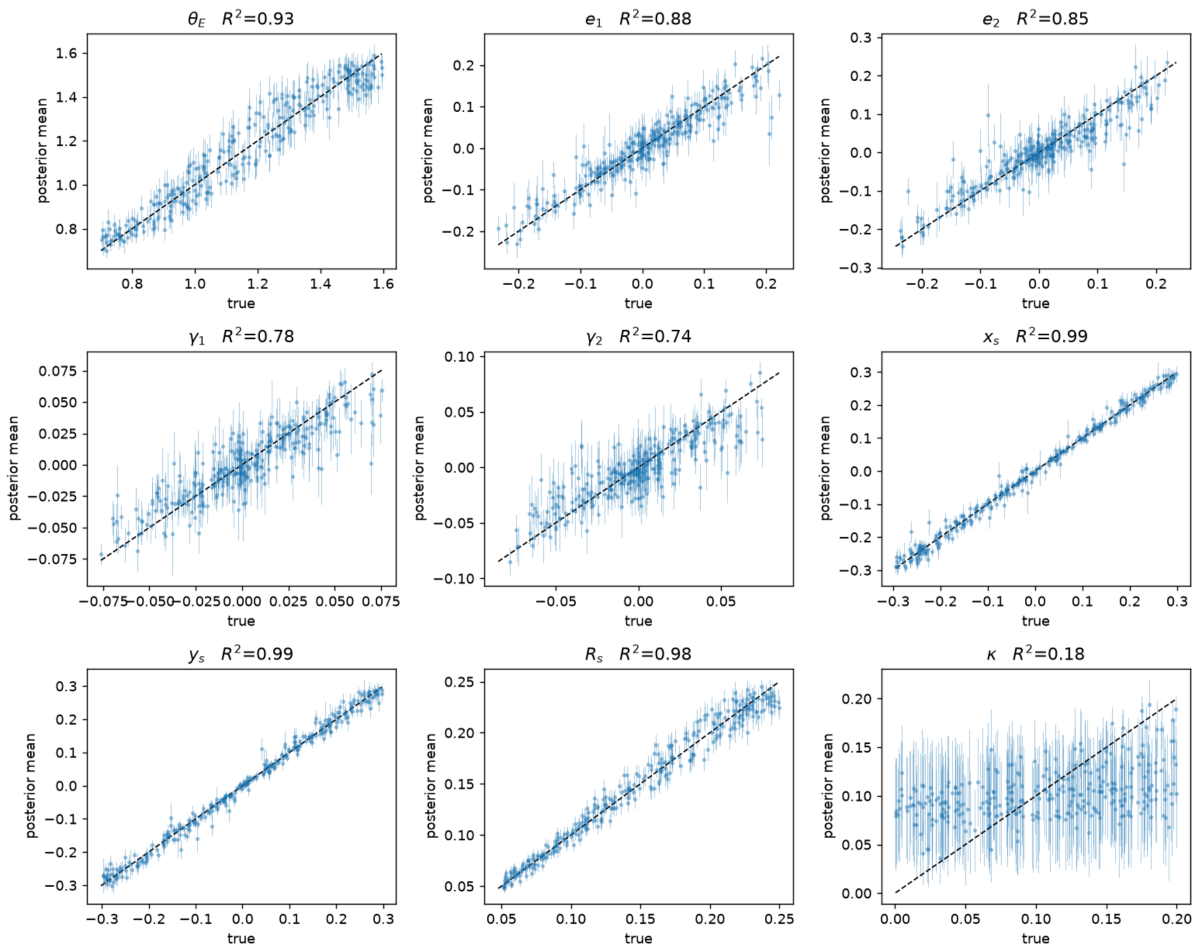

Figure 5: Recovery for the 9-parameter model including external convergence κ. The κ posterior (bottom right) stays near the prior mean with near-prior-width uncertainties ($R^2 = 0.18$), and θ_E recovery (top left) degrades from $R^2 = 1.00$ to 0.93 relative to Figure 2—the mass-sheet degeneracy in action.

That an off-the-shelf neural posterior estimator recovers this degeneracy structure—without being told about it—is a meaningful validation of the SBI approach: the network reports joint posterior *geometry*, not just marginal point estimates. In practice, breaking this degeneracy requires external information (stellar kinematics of the lens, time delays, or statistical priors on the line-of-sight environment), none of which is present in a single image.

5 Discussion and Limitations

Effect of signal-to-noise (a controlled comparison). The first iteration of this project used an untuned source amplitude giving median peak SNR ≈ 4; after tuning the amplitude ($20 \to 150$) into the assignment's 10–30 target, every parameter's recovery improved (Table 2), most dramatically the weakly-imprinting ones (shear: R^2 $0.06 \to 0.65$; ellipticity: $\approx 0.5 \to \approx 0.87$). The mass-sheet degeneracy conclusions were unchanged in kind—as expected, since they are information-theoretic rather than noise-driven—but the degeneracy ridge sharpened (corr $-0.83 \to -0.93$).

Calibration. At $N_{\text{train}} = 8{,}000$ the model overfit (validation loss rising after epoch $\sim$20) and its posteriors were overconfident; at $N_{\text{train}} = 20{,}000$ neither problem remains, with SBC showing calibrated to mildly *under*confident posteriors. If tighter calibration were required, the remaining slack could be addressed with a larger validation-selected model or more training data (50,000 simulations remain feasible); the current conservative posteriors are already safe for downstream use.

Model scope. The forward model fixes several quantities (lens centroid, source amplitude and profile shape, PSF) that would be uncertain for real observations, and simulates only the lensed source (no lens-galaxy light). Extending $\boldsymbol{\theta}$ with these nuisance parameters, or marginalizing over them in the simulator, is straightforward within the same pipeline and is the natural next step toward application to survey data.

Computational profile. Training the 20,000-sample model takes 50 minutes on an RTX 3060 ($\approx$ 117 ms/step, vs. $\approx$ 500 ms/step on CPU); dataset simulation takes $\sim$10 minutes CPU-parallelized. Once trained, the amortized posterior costs milliseconds per system (sampling 2,000 draws for all 300 test systems took 0.3 s)—the property that makes this approach attractive for the 10^4–10^5 lens samples expected from upcoming surveys.

6 Conclusion

We built and validated an end-to-end amortized SBI pipeline for strong-lens parameter inference. From single noisy 64×64 images at realistic SNR, the BayesFlow posterior estimator recovers the Einstein radius, source position, and source size near-perfectly ($R^2 = 0.99$–1.00), the lens ellipticity accurately ($R^2 = 0.84$–0.89), and even the weak external shear meaningfully ($R^2 = 0.65$–0.80). Simulation-based calibration shows statistically calibrated posteriors for the weakly-constrained parameters and mildly conservative (underconfident) ones for the best-constrained—the safe direction of miscalibration. A controlled comparison against an earlier low-SNR, smaller-dataset run demonstrates both the information value of SNR (shear recovery went from unusable to good) and the calibration value of training-set size (overconfident U-shaped SBC histograms became flat). When external convergence is included, the inferred posteriors exhibit the mass-sheet degeneracy exactly as theory predicts: only 17% contraction of κ relative to the prior, a strong κ–θ_E anti-correlation (-0.93), and a corresponding degradation of θ_E recovery from $R^2 = 1.00$ to 0.93 while all other parameters are untouched. The complete pipeline—simulation, training, and evaluation—runs in about an hour per model variant on a single consumer GPU, after which inference on new systems is effectively instantaneous.

References

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